Exact Finite-Field Moment Reductions for Arithmetic Quantum Cat Maps and the Remaining BGS Frontier

Maksim Churilov

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Abstract

We isolate an exact finite-field object controlling a character-averaged fourth moment in an arithmetic quantum cat-map model. The correct object is a weighted pair-key distribution $B(t)$; an earlier Möbius-fiber assembly is shown to be a different finite object and is not used. We prove exact pair-key, counting, character-orthogonality and Fourier-variance identities. The mixed part of B is transformed by a Cayley map into an additive-energy problem. For each fixed odd prime subgroup order K, characteristic-zero collisions are classified by a finite cyclotomic matching problem and finite-field exceptions are detected by explicit resultants. The audited sets for $K = 11, 13$ are stated, together with the exact exceptional control $(N, K) = (461, 23)$. The remaining off/off term is organized by a non-isotrivial elliptic pencil. We state the precise algebraic and sheaf-theoretic obligations that still prevent promotion of the resulting $K = o(N^{1/4})$ fourth-moment estimate to an unconditional theorem. Finite GOE-like spacing and rigidity data are recorded only as numerical evidence. No microscopic BGS eigenphase law is claimed.

1 Scope and evidence boundary

The Bohigas–Giannoni–Schmit principle predicts random-matrix local spectral statistics for suitable quantizations of classically chaotic systems after exact symmetries are removed. Arithmetic cat maps are delicate because their finite-field and Hecke structures can produce exact degeneracies and long-range arithmetic dependence. A matrix-element moment identity, a quantum-ergodicity theorem, or a finite Wigner–Dyson fit does not by itself imply a microscopic GOE/GUE eigenphase law.

The purpose of this note is therefore narrower and testable:

1. identify the exact finite-field fourth-moment object;
2. separate its deterministic, mixed and genuinely two-parameter pieces;
3. prove the exact finite reductions;
4. isolate the remaining geometric correlation problem without importing a local-law theorem before its hypotheses are available.

We use the following evidence labels throughout. An *exact identity* is a finite algebraic/combinatorial statement. A *finite certificate* is a statement at explicitly listed finite designs, including machine-checked Lean calculations. A *conditional proposition* records a precise implication whose load-bearing hypotheses remain to be certified. Numerical spectral calculations are never used as proofs.

2 Finite-field setup and pair keys

Let N be an odd prime and

$$H \le \mathbf{F}_N^\times, \qquad |H| = K, \qquad m = \frac{N-1}{K}.$$

Let $S = H^\perp$ denote the annihilator subgroup of multiplicative characters. For $x, y \in \mathbf{F}_N \setminus \{0,1\}$ define

$$u = \frac{x}{y}, \qquad v = \frac{x-1}{y-1}, \qquad d = x - y.$$

Proposition 2.1 (Pair-key parametrization)**.** *If $x \neq y$, then $(x,y) \mapsto (u,v)$ is a bijection onto*

$$D_{\text{off}} = \{(u,v) \in (\mathbf{F}_N^\times \setminus \{1\})^2 : u \neq v\},$$

with inverse

$$y = \frac{1-v}{u-v}, \qquad x = \frac{u(1-v)}{u-v},$$

and

$$d(u,v) = \frac{(u-1)(v-1)}{v-u}.$$

On the diagonal $x = y$, all $N-2$ admissible pairs have key $(1,1)$ and difference 0.

Proof. From $u = x/y$ and $v = (x-1)/(y-1)$ one has $x = uy$ and $uy - 1 = v(y-1)$. Hence $(u-v)y = 1-v$, which gives the inverse when $u \neq v$. Substitution gives the displayed $d(u,v)$. Conversely, for $u, v \in \mathbf{F}_N^\times \setminus \{1\}$ with $u \neq v$, the formulas give $x, y \notin \{0,1\}$ and recover the prescribed key. If $x = y$, both ratios are 1 and $d = 0$; the admissible diagonal has $N-2$ points. □

The diagonal multiplicity is load-bearing. Treating $(1,1)$ as an ordinary unit-weight point changes the fourth-moment object.

3 Exact second- and fourth-moment reductions

A collapsed key record is

$$r = (u_r, v_r, d_r, n_r),$$

where n_r is the multiplicity. Define

$$B(t) = \sum_{r,s} n_r n_s \, \mathbf{1}_H(u_r u_s) \, \mathbf{1}_H(v_r v_s) \, \mathbf{1}_{d_r + d_s = t}.$$

Lemma 3.1 ($k = 2$ constrained count)**.** *The number of off-diagonal keys in $(H \setminus \{1\})^2$ is*

$$(K-1)(K-2).$$

Proof. There are $(K-1)^2$ ordered pairs with both entries nontrivial and $K-1$ diagonal pairs $u = v$ to remove. □

Theorem 3.2 (Exact $k = 4$ total)**.** *For every odd prime N and every multiplicative subgroup H of order K,*

$$\sum_t B(t) = (K^2+1)N^2 - (6K^2+3K)N + 13K^2 - 3K.$$

Proof. Split the ordered coupling into diagonal/diagonal, diagonal/off, off/diagonal and off/off classes. The first contributes $(N-2)^2$. Each mixed class contributes

$$(N-2)(K-1)(K-2).$$

For an off-diagonal key (u, v) a partner has the form $(h_1/u, h_2/v)$, $(h_1, h_2) \in H^2$. Failure to remain in D_{off} occurs through $h_1 = u$, $h_2 = v$, or $h_1 v = h_2 u$. Inclusion–exclusion gives

$$V_{\text{off}} = K^2(N-2)(N-3) - 3K(K-1)(N-3) + 2(K-1)(K-2).$$

Adding the four ordered classes and simplifying yields the formula. □

For $\chi, \psi \in S$ put

$$A_{\chi,\psi}(d) = \sum_{r:d_r=d} n_r \chi(u_r)\psi(v_r).$$

Theorem 3.3 (Character collapse)**.** *For every $t \in \mathbf{F}_N$,*

$$\sum_{\chi,\psi \in S} (A_{\chi,\psi} * A_{\chi,\psi})(t) = m^2 B(t).$$

Proof. Expand the convolution over ordered key records. The sum over $\chi \in S$ is m precisely when $u_r u_s \in H$ and is zero otherwise; the same holds independently for ψ and the v-constraint. □

For the twisted Jacobi sums

$$J(\chi, \psi; q) = \sum_{x \in \mathbf{F}_N \setminus \{0,1\}} \overline{\chi(x)}\, \psi(x-1) e_N(qx)\,,$$

the same expansion gives

$$\sum_{\chi,\psi \in S} |J(\chi, \psi; q)|^4 = m^2 \Sigma_4(q), \qquad \Sigma_4(q) = \sum_t B(t) e_N(qt)\,.$$

4 Exact Fourier mean and variance

Let

$$V = \sum_t B(t), \qquad Z = B(0).$$

Theorem 4.1 (Finite Fourier duality)**.** *Averaging over nonzero additive frequencies,*

$$\mathbf{E}_{q \neq 0} \Sigma_4(q) = \frac{NZ - V}{N-1},$$

and

$$\operatorname{Var}_{q \neq 0} \Sigma_4(q) = N \operatorname{Var}_{t \neq 0} B(t).$$

Proof. Use

$$\sum_{q \neq 0} e_N(qt) = \begin{cases} N-1, & t = 0, \\ -1, & t \neq 0, \end{cases}$$

for the mean. For the second centered moment, expand the Fourier transform, apply the same complete orthogonality relation to $t - s$, and simplify the nonzero-frequency mean term. No asymptotic input is used. □

Consequently,

$$\operatorname{std}_{q \neq 0} \Sigma_4(q) = O(NK)$$

is exactly equivalent to

$$\operatorname{std}_{t \neq 0} B(t) = O(K\sqrt{N}).$$

5 Deterministic, mixed and off/off decomposition

Let

$$C_H(t) = \#\{(u,v) \in (H \setminus \{1\})^2 : u \neq v,\ d(u,v) = t\}.$$

Writing $O(t)$ for the pure off/off contribution gives the exact decomposition

$$B(t) = O(t) + 2(N-2)C_H(t) + (N-2)^2 \mathbf{1}_{t=0}.$$

Thus the variance problem separates into a deterministic spike, a mixed one-dimensional energy and a genuinely two-parameter off/off family.

6 Cayley transform and exact mixed energy

For $h \neq 1$ define

$$c(h) = \frac{h+1}{h-1}.$$

Lemma 6.1 (Cayley difference identity)**.** *For $u, v \neq 1$, $u \neq v$,*

$$d(u,v) = \frac{2}{c(u) - c(v)}.$$

Moreover,

$$c(h^{-1}) = -c(h).$$

Hence, up to the bijection $\Delta \mapsto 2/\Delta$ on $\mathbf{F}_N^\times$, C_H is the nonzero additive-difference representation function of

$$A = c(H \setminus \{1\}).$$

For odd K, $0 \notin A$ and $A = -A$. The central symmetry forces the baseline collision energy

$$E_{\text{forced}} = (K-1)(2K-5).$$

For odd prime K, clearing denominators in

$$c(\zeta^i) - c(\zeta^j) = c(\zeta^r) - c(\zeta^s)$$

gives a sparse cyclotomic relation. The current exact computer-assisted classification checks all $6! = 720$ matchings. Over characteristic zero the admissible kernels reduce to the identity and inversion-swap families. Finite-field reduction can create additional collisions.

For a fixed odd prime K, let E_K be the finite set of rational primes $N \equiv 1 \pmod K$ dividing the relevant nonzero cyclotomic resultants.

Proposition 6.2 (Fixed-K mixed energy)**.** *If $N \equiv 1 \pmod K$ is prime and $N \notin E_K$, then*

$$\sum_t C_H(t)^2 = (K-1)(2K-5).$$

The exact resultant gate currently fully enumerates

$$E_{11} = \{23, 67, 89\}, \qquad E_{13} = \{53, 79, 131, 157, 313\}.$$

The qualification "outside E_K" is essential. A fresh independent exact calculation at

$$(N, K) = (461, 23)$$

gives

$$\sum_t C_H(t)^2 = 1190, \qquad (K-1)(2K-5) = 902.$$

Thus this design is exceptional for the forced-energy formula.

Outside the exceptional reductions the mixed term $2(N-2)C_H(t)$ has standard deviation $O(K\sqrt{N})$, which is precisely the global target scale.

7 The off/off elliptic pencil

Use Cayley coordinates

$$a = c(u), \qquad b = c(v), \qquad \alpha = c(h_1), \qquad \beta = c(h_2),$$

and

$$T_\alpha(a) = \frac{\alpha a - 1}{a - \alpha}.$$

A fixed parameter pair is governed by

$$\frac{2}{a-b} + \frac{2}{T_\alpha(a) - T_\beta(b)} = t.$$

After clearing denominators the generic fiber is a plane cubic. Solving quadratically in one variable gives a binary-quartic double-cover model.

Current symbolic work supports the following structural picture:

- the generic j-map is nonconstant, so the family is non-isotrivial;
- there are two special $O(K)$ parameter strata corresponding to $\alpha = \beta$ and $\alpha + \beta = 0$;
- an involution $(h_1, h_2) \mapsto (h_2^{-1}, h_1^{-1})$ preserves the generic same-base j-data;
- the order-four bad fiber occurs at the same Cayley-difference parameter measured by C_H and factors into two lines.

A broader algebraic draft in the research record derives a degree-12 j-map, cusp-width restrictions on possible cyclic isogenies, and affine boundary bookkeeping. These calculations are valuable, but the present note does not promote them to a theorem until the following obligations are closed.

Problem 7.1 (Algebraic family certificate)**.** *Produce a publication-grade projective/Weierstrass model with:*

1. *exact discriminant and j-map;*
2. *complete exceptional-parameter list;*
3. *twist and same-j classification;*
4. *separable and inseparable isogeny exclusions after reduction;*
5. *explicit treatment of all deleted affine boundary components.*

Problem 7.2 (Trace-sheaf correlation input)**.** *Identify one precise source theorem and check, for the actual family: purity, geometric irreducibility/monodromy, conductor and ramification bounds, duals/twists, deleted points and geometric non-isomorphism of every off-diagonal parameter pair to which square-root correlation is applied.*

Proposition 7.3 (Conditional restricted M1 estimate)**.** *Assume the two preceding obligations with constants uniform in the allowed parameters. Then*

$$\mathrm{Var}_{t \neq 0}\, O(t) \ll K^2 N + K^4 N^{1/2} + K^4.$$

Consequently, outside the relevant fixed-K cyclotomic exceptions and in the first regime

$$K = o(N^{1/4}),$$

one obtains

$$\mathrm{Var}_{t \neq 0}\, B(t) = O(K^2 N), \qquad \mathrm{std}_{q \neq 0}\, \Sigma_4(q) = O(NK).$$

Remark 7.4. *This proposition is conditional. It is intentionally not labelled as an unconditional theorem in the repository.*

8 Finite certificates and reproducibility anchors

Independent finite constructions currently agree on

N	K	$\sum_t B(t)$	$B(0)$	$\sum_t B(t)^2$
23	11	48 621	2 609	103 156 981
53	13	423 879	12 063	3 410 502 105
199	11	4 681 821	96 641	115 589 503 389
461	23	111 147 915	641 799	26 960 554 630 421

with Python and an independent C++ implementation used where independence is scientifically useful.

The Lean project contains finite certificates for:

- the $k = 2$ counting/bijection layer at an audited design;
- the $k = 4$ variety/key multiplicities at an audited design;
- torus level-set/orbit facts at $N = 199$;
- the scale-law identity at the audited $N = 199$ design;
- the exceptional Cayley energy at $(461, 23)$.

The file `OpLocal.lean` is a specification with admitted obligations, not a completed theorem.

9 Recovered earlier foundations

The broader research record predates the pair-key repair. It contains substantial work on the symmetry class, quadratic trace identities, Kloosterman/character expansions, finite moments and shape decompositions, torus level sets and twisted Jacobi formulas. Those results must not be lost when the current M1 route is rewritten.

At the same time, an old label "PROVEN" is not by itself sufficient evidence. The present reconstruction therefore separates:

1. pieces re-certified by current finite Lean/Python/C++ gates;
2. analytically plausible historical theorem statements awaiting a fresh line-by-line audit;
3. numerical evidence;
4. permanently retracted routes.

This is a preservation rule, not a weakening of the scientific objective.

10 Spectral evidence and the open BGS bridge

At retained parity-reduced designs, nearest-neighbor spacings are compatible with the GOE Wigner surmise and number-variance calculations display comparable finite-sample rigidity. These are motivation and falsification tools only.

A historical $N = 95419$ script compared a gap-ratio statistic with the ordinary spacing CDF; its named KS verdict is therefore quarantined. This does not invalidate the separately retained spacing calculations at other designs.

Even an unconditional restricted M1 theorem would not finish BGS. The next load-bearing question is to identify a comparison mechanism for the structured arithmetic matrix family at microscopic scale. A valid route must specify the symmetry-reduced ensemble, dependence structure, resolvent or correlation observable, limiting regime and quantitative error. Importing a Wigner local law without those checks would silently change the problem.

11 Permanent retractions

The active argument does not use:

- the historical Möbius-fiber/M4D proof of M1;
- the claim that finite Catalan moments plus an Erdős–Schlein–Yau citation automatically imply BGS;
- the unresolved higher-word "certificate";
- the failed Hecke-centering Julia series;
- the historical `cdense_transfer.jl` Perron bracket;
- the historical `sol6_closed_form.jl` closed-form claim;
- extrapolation from rank-2 diagonal/coset calculations to a general $\mathrm{Sp}(2g)$ theorem.

Selected references

1. P. Kurlberg and Z. Rudnick, *On the distribution of matrix elements for the quantum cat map*, Ann. of Math. 161 (2005), 489–507, DOI 10.4007/annals.2005.161.489.
2. A. Gamburd, J. Lafferty and D. Rockmore, *Eigenvalue spacings for quantized cat maps*, J. Phys. A 36 (2003), DOI 10.1088/0305-4470/36/12/336.
3. E. Fouvry, E. Kowalski and P. Michel, *A study in sums of products*, Phil. Trans. R. Soc. A 373 (2015), DOI 10.1098/rsta.2014.0309.
4. E. Fouvry, E. Kowalski, P. Michel, C. S. Raju, J. Rivat and K. Soundararajan, *On short sums of trace functions*, Ann. Inst. Fourier 67 (2017), DOI 10.5802/aif.3087.
5. W. Sawin, A. Forey, J. Fresan and E. Kowalski, *Quantitative sheaf theory*, J. Amer. Math. Soc. 36 (2023), 653–726, DOI 10.1090/jams/1008.
6. P. Michel, *Rang moyen de familles de courbes elliptiques et lois de Sato–Tate*, Monatsh. Math. 120 (1995), 127–136.
7. Y. G. Zarhin, *Homomorphisms of abelian varieties over geometric fields of finite characteristic*, arXiv:1012.5083.